Linux, Web Server & Cloud Basics Assignment

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1. Difference between Containers and Blob Storage

Feature	Containers	Blob Storage
Purpose	Package and run applications in isolated environments.	Store large amounts of unstructured data.
Contains	Application, libraries, dependencies and configuration.	Files/objects such as images, videos, backups and documents.
Main use	Application deployment and execution.	Cloud data storage and retrieval.
Example	Docker container	Azure Blob Storage container/object

2. Commands for installation of a web server

Apache Web Server on Ubuntu/Debian:

```
sudo apt update
sudo apt install apache2
sudo systemctl start apache2
sudo systemctl enable apache2
sudo systemctl status apache2
```

Apache Web Server on RHEL/CentOS:

```
sudo dnf install httpd
sudo systemctl start httpd
sudo systemctl enable httpd
sudo systemctl status httpd
```

3. What are the types of web servers?

Common web server software includes:

- **Apache HTTP Server** – widely used open-source web server.
- **Nginx** – high-performance web server and reverse proxy.
- **Microsoft IIS** – web server provided by Microsoft for Windows Server.
- **LiteSpeed** – high-performance web server used for many hosting environments.

4. What is RHEL?

RHEL stands for Red Hat Enterprise Linux. It is a commercial Linux distribution developed by Red Hat for enterprise use. It is commonly used for servers, cloud systems, databases, networking, and business applications. RHEL is known for enterprise support, security features, stability, and long-term maintenance.

5. Difference between Linux, CentOS and Ubuntu

Item	Linux	CentOS	Ubuntu
Meaning	A family/kernel ecosystem of open-source Unix-like operating systems.	A Linux distribution historically derived from RHEL sources.	A Linux distribution based on Debian.
Typical use	Servers, desktops, embedded systems and more.	Traditionally common in servers and enterprise environments.	Common on desktops, servers, cloud and development systems.

Item	Linux	CentOS	Ubuntu
Package tools	Depends on the distribution.	Historically RPM/YUM; newer CentOS versions use DNF.	APT/Dpkg.
Organization	Linux kernel/community ecosystem.	CentOS project; current CentOS Stream is sponsored by Red Hat.	Developed by Canonical and the Ubuntu community.

6. What is static web hosting?

Static web hosting is a hosting method that serves fixed website files – HTML, CSS, JavaScript, images and fonts – directly to the visitor's browser exactly as they are stored. There is no server-side processing or database; every visitor receives the same content until the files are updated. Any interactivity is handled in the browser by client-side JavaScript or by calling external APIs.

Key features:

• **Fast** – files are served as-is and can be cached or delivered through a CDN.
• **Low cost** – needs only simple storage and bandwidth, not application servers.
• **Secure** – no server-side code or database reduces the attack surface.
• **Scalable** – easily handles high traffic since no processing is required.

Examples: Azure Blob Storage static website hosting, Azure Static Web Apps, AWS S3, GitHub Pages and Netlify. It suits portfolios, landing pages, documentation and blogs, but not sites that need server-side logic such as user logins with a backend database (those need dynamic hosting).

Note: Commands can vary slightly depending on the Linux distribution and its version. The examples above use common Ubuntu/Debian and RHEL/CentOS commands.